AGRONOMY AI

Your Intelligent Voice-First Agricultural Assistant

GREAT INDIAN AI INTERNSHIP CHALLENGE – GNANI AI

GitHub: github.com/nani3467/Plant-Disease-Detection

⚠️ SECTION 1: THE PROBLEM

- **Early Disease Detection:** Farmers struggle to identify crop infections early, risking severe yield loss and costly late interventions.
- **Complex Crop Selection:** Selecting optimal crops requires intricate soil NPK balancing, pH understanding, and climate matching.
- **Information Barriers:** Fragmented access to scientific ICAR agronomy guidelines leaves farmers dependent on hearsay.
- **Digital Literacy & Language Gap:** Traditional typing-heavy apps fail rural farmers due to script and literacy barriers.

🌱 SECTION 2: OUR SOLUTION

- **Voice-First Accessibility:** Hands-free voice interface in native Indian languages powered by Gnani AI speech engines.
- **AI Vision Diagnostics:** Instant leaf disease detection across 38 classes via deep learning MobileNetV2.
- **Intelligent Crop Recommendation:** Data-driven soil and climate matching for 22 crops using trained Random Forest ML.
- **ICAR-Grounded Guidance:** 5-part actionable reports covering organic remedies, approved chemicals, dosages, and safety PPE.

🎙️ SECTION 3: GNANI AI INTEGRATION — CORE SPEECH ENGINE & ACCESSIBILITY HERO

Verified Backend Vachana REST API Integration

GNANI PRISMA | SPEECH-TO-TEXT

Model: prisma-v3.0

- **Spoken Query Conversion:** Converts farmer's raw microphone audio (WAV/WebM) into high-accuracy text at `api.vachana.ai/stt/v3`.
- **Multilingual Speech Recognition:** Built-in Indian vernacular support for Telugu (`te-IN`), Hindi (`hi-IN`), and English (`en-IN`) with auto-detection.
- **Typing-Free Navigation:** Empowers farmers to issue hands-free voice commands, navigate pages, and trigger ML workflows without typing.

GNANI TIMBRE | TEXT-TO-SPEECH

Model: timbre-v3.0

- **Spoken Agricultural Guidance:** Synthesizes AI diagnosis & advice into natural, human-sounding speech via `api.vachana.ai/api/v1/tts/inference`.
- **Native Vernacular Voices:** Dynamic voice allocation—Lakshmi (Telugu), Shivani (Hindi), and Avinash (Indian English).
- **Acoustic Optimization:** Converts 24kHz linear PCM to valid RIFF WAV with broadcast loudness normalization (-14 dBFS) & HTML5 repair.

1. INPUT Farmer Speaks Voice Query → **2. GNANI PRISMA STT** Transcribes Audio to Vernacular Text → **3. AGRONOMY AI & GEMINI** Parses Intent & Formulates Agronomic Advice → **4. GNANI TIMBRE TTS** Synthesizes High-Quality Spoken Audio → **5. OUTPUT** Farmer Hears Actionable Guidance

⚡ SECTION 4: KEY PLATFORM FEATURES

🪲 Plant Disease Detection — VERIFIED ML

Upload leaf photos to detect 38 plant disease classes across 14+ crops via PyTorch MobileNetV2 with non-plant heuristic filtering.

🌾 Crop Recommendation — VERIFIED ML

Recommends optimal crop out of 22 varieties based on soil N-P-K, pH, temperature, humidity, and rainfall using Random Forest (99.5% accuracy).

📖 Botanical & Crop Guides — VERIFIED

Comprehensive database of 22 crop manuals and 38 disease dossiers containing symptoms, pathogen info, and seasonal cultivation requirements.

🎙️ Voice-First Assistant — VERIFIED

Floating Voice OS assistant integrating Gnani Prisma STT and Timbre TTS for hands-free audio recording, intent execution, and voice playback.

🌐 Multilingual Fluency — VERIFIED

Full end-to-end support for English, Telugu (తెలుగు), and Hindi (हिन्दी) across voice recognition, reasoning, and speech synthesis.

🧪 Fertilizer & IoT Scouting — IN DEVELOPMENT

Interactive deficiency guide live; automated IoT telemetry sensors and drone aerial multispectral mapping currently under active development.

🛠️ SECTION 5: VERIFIED TECHNOLOGY STACK

Speech AI	**Gnani AI Prisma v3.0** (STT) & **Timbre v3.0** (TTS) via Vachana API
Frontend	Next.js 14.0.4, React 18, TypeScript 5, Tailwind CSS 3.3, Axios, jsPDF
Backend	Django 5.2.5, Django REST Framework 3.15.2, Python 3.11, Gunicorn
Database	MongoDB Atlas / Community via `django-mongodb-backend`
ML Models	PyTorch MobileNetV2 (38 diseases), scikit-learn Random Forest (22 crops)
Reasoning	Google Gemini (`gemini-2.5-flash` via `google-genai`) + Local ICAR Rules

🎯 SECTION 6: IMPACT & FUTURE SCOPE

- **Bridges Rural Digital Divide:** Enables farmers of any literacy level to access advanced AI diagnostics using natural spoken dialects.
- **Prevents Catastrophic Crop Losses:** Rapid leaf-level diagnosis catches fungal and bacterial blights before widespread infestation.
- **Optimizes Soil Yields:** Scientific NPK-based crop selection maximizes harvest returns and prevents soil nutrient depletion.
- **Future Roadmap:** Real-time IoT soil telemetry integration, offline-capable mobile PWA, and drone aerial multispectral crop scouting.